THE NORM OF THE PARTIAL SUM OPERATOR ON THE HARDY SPACE $\mathcal{H}^1$ OF DIRICHLET SERIES

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ABSTRACT. Let S_N denote the Nth partial sum operator for Dirichlet series. We show that the norm of S_N on the Hardy space $\mathcal{H}^1$ of Dirichlet series is of order $\log N/\log\log N$. The upper bound is due to Bondarenko, Brevig, Saksman and Seip, and the previously known lower bound was of order $\log\log N$. This answers the case $p = 1$ of a problem of Saksman and Seip, up to the value of the leading constant. The lower bound comes from an entropy inequality for half-space projections on finite-dimensional tori, which is the ordered-group form of Zygmund's $L\log L$ converse, applied to a product of $d \asymp \log N/\log\log N$ independent factors.

1. INTRODUCTION

Let $\mathbb{T}^\infty$ be the infinite torus with its normalized Haar measure m, and let $p_1 < p_2 < \cdots$ be the primes. If $n = \prod_j p_j^{\kappa_j}$, the Bohr lift of the Dirichlet polynomial $f(s) = \sum_{n\le N} a_n n^{-s}$ is the polynomial $\mathcal{B}f(z) = \sum_{n\le N} a_n z^{\kappa(n)}$ on $\mathbb{T}^\infty$. For $0 < p < \infty$ the Hardy space $\mathcal{H}^p$ is the completion of the Dirichlet polynomials in the norm

$$\|f\|_{\mathcal{H}^p} = \left(\int_{\mathbb{T}^\infty} |\mathcal{B}f(z)|^p \,\mathrm{d}m(z)\right)^{1/p}.$$

For $N \ge 1$ the partial sum operator is

$$S_N\left(\sum_{n=1}^{\infty} a_n n^{-s}\right) = \sum_{n=1}^{N} a_n n^{-s}.$$

We write $\|S_N\|_p$ for its norm on $\mathcal{H}^p$, that is, the supremum of $\|S_N f\|_{\mathcal{H}^p}$ over Dirichlet polynomials f with $\|f\|_{\mathcal{H}^p} = 1$.

The dual group of $\mathbb{T}^\infty$ is identified with the multiplicative group of positive rationals, ordered by size, and S_N keeps the frequencies which are at most N in this order. Helson [3] showed that such projections are bounded on L^p for $1 < p < \infty$, so that $\|S_N\|_p$ is bounded independently of N in this range. For $p = 1$ this fails. The one-variable Dirichlet series $\sum_k a_k 2^{-ks}$ give the

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classical lower bound $\|S_N\|_1 \gtrsim \log\log N$, and Bondarenko, Brevig, Saksman and Seip [1, Theorem 5.2] proved that

$$\log\log N \lesssim \|S_N\|_1 \lesssim \frac{\log N}{\log\log N}. \tag{1}$$

Konyagin, Queffélec, Saksman and Seip [4, Section 5] later recorded the same gap. They also noted that the argument of [1] adapts to Dirichlet series over restricted sets of primes, where it gives a bound in terms of the maximal order of the divisor function; for a fixed finite set of primes this bound is of order $\log\log N$. Saksman and Seip [5, Problem 4.5] asked the following.

Problem 1 (Saksman–Seip)**.** *Determine the precise asymptotic growth of the norm of the partial sum operator* $S_N\colon \sum_{n=1}^{\infty} a_n n^{-s} \mapsto \sum_{n=1}^{N} a_n n^{-s}$ *when* $N \to \infty$ *for* $p = 1$ *(or more generally, for* $p \le 1$ *or* $p = \infty$*).*

The main purpose of the present paper is to show that the upper bound in (1) gives the correct order.

Theorem 1. *There are absolute constants* $c > 0$ *and* C *such that, for every integer* $N \ge 16$*,*

$$c\frac{\log N}{\log\log N} \le \|S_N\|_1 \le C\frac{\log N}{\log\log N}.$$

The upper bound is [1, Theorem 5.2], and the content of Theorem 1 is the lower bound. In particular, this answers the case $p = 1$ of the problem of Saksman and Seip up to the value of the leading constant. To the best of our knowledge, the lower bound is new.

Let us briefly describe our approach. On Dirichlet polynomials over the first d primes, a partial sum with a suitable shift becomes the projection P_λ on $\mathbb{T}^d$ which keeps the frequencies γ with $\lambda \cdot \gamma \le 0$, where $\lambda_j = \log p_j$. For a nonnegative function u of mean one, the imaginary part of $P_\lambda u$ controls the entropy of u,

$$\int u \log u \, \mathrm{d}m \le \pi \|P_\lambda u\|_{L^1(\mathbb{T}^d)}.$$

This is the ordered-group form of the classical argument behind Zygmund's converse theorem, that a nonnegative function with integrable conjugate function belongs to $L\log L$ [6, Chapter VII, Section 2]. Entropy is additive for products, so $u(z) = \prod_{j\le d}(1 + \cos\theta_j)$ gives a lower bound of order d, and d can be taken of order $\log N/\log\log N$. In contrast to the lower bound in (1), which uses one prime, the extremal example uses a number of primes which grows with N. The same product over the first d primes, and the same passage from primorials to general N, appear in the proof of the lower bound for $0 < p < 1$ in [1, Theorem 5.3]. There the lower bound comes from a single coefficient, and for $p = 1$ this coefficient equals 1. The new ingredient is the entropy inequality, which bounds the L^1 norm of the projection from below.

Organization. The present paper is organized into four additional sections. In Section 2 we collect the standard facts that we need. In Section 3 we prove the entropy inequality for half-space projections. In Section 4 we apply it to a product over the first d primes and prove Theorem 1. Section 5 contains remarks on the leading constant and on the other values of p.

2. Preliminaries

Fix $d \geq 1$, let m also denote the normalized Haar measure of $\mathbb{T}^d$, and write $z_j = e^{i\theta_j}$. A trigonometric polynomial on $\mathbb{T}^d$ is a finite sum $u(z) = \sum_\gamma \widehat{u}(\gamma) z^\gamma$ with $\gamma \in \mathbb{Z}^d$. Put $\lambda = (\log p_1, \ldots, \log p_d)$. The frequency $\gamma \in \mathbb{Z}^d$ corresponds to the rational number $\prod_j p_j^{\gamma_j}$, whose logarithm is $\lambda \cdot \gamma$. By unique factorization,

$$\lambda \cdot \gamma \neq 0 \qquad \text{for every } \gamma \in \mathbb{Z}^d \setminus \{0\}. \tag{2}$$

For a vector $\lambda \in \mathbb{R}^d$ which satisfies (2) we define

$$P_\lambda u = \sum_{\lambda \cdot \gamma \leq 0} \widehat{u}(\gamma) z^\gamma.$$

We will use two elementary facts. First, if $b \geq 1$ is an integer, then the Bohr lift of $b^{-s} f$ is $z^{\kappa(b)} \mathcal{B} f$, so multiplication by b^{-s} preserves the norm of $\mathcal{H}^1$. Second, we need Chebyshev's lower bound for the number of primes, in the form

$$\log \prod_{j \leq d} p_j \leq C_0 d \log(d+1), \qquad d \geq 1, \tag{3}$$

for an absolute constant C_0. For completeness we recall the argument. If $p \leq 2n$ is prime, then the exponent of p in $\binom{2n}{n}$ is $\sum_{k \geq 1} (\lfloor 2n/p^k \rfloor - 2\lfloor n/p^k \rfloor)$, a sum of at most $\log(2n)/\log p$ terms each equal to 0 or 1. Hence each such prime contributes at most $\log(2n)$ to $\log \binom{2n}{n}$. Since $\binom{2n}{n} \geq 4^n/(2n+1)$, the number $\pi(x)$ of primes $p \leq x$ satisfies $\pi(x) \gtrsim x/\log x$ for $x \geq 2$. Applying this at $x = p_d$ we get $p_d \lesssim d \log(d+1)$, and (3) follows from $\log \prod_{j \leq d} p_j \leq d \log p_d$.

3. An entropy inequality for half-space projections

The next result is the only place where we use that the input is nonnegative. It bounds the entropy of u by the size of the imaginary part of $P_\lambda u$. Here $0 \log 0 = 0$.

Lemma 1. *Suppose that $\lambda \in \mathbb{R}^d$ satisfies (2) and that $u \geq 0$ is a real trigonometric polynomial on $\mathbb{T}^d$ with $\int u \, \mathrm{d}m = 1$. Put $G = 2P_\lambda u - 1$. Then $\operatorname{Re} G = u$, and $v = \operatorname{Im} G$ satisfies*

$$\int_{\mathbb{T}^d} u \log u \, \mathrm{d}m \leq \frac{\pi}{2} \|v\|_{L^1(\mathbb{T}^d)}.$$

Consequently, $\|P_\lambda u\|_{L^1(\mathbb{T}^d)} \geq \pi^{-1} \int u \log u \, \mathrm{d}m$.

Proof. Since u is real, $\widehat{u}(-\gamma) = \overline{\widehat{u}(\gamma)}$. By (2), every $\gamma \neq 0$ lies in exactly one of the two open half-spaces $\lambda \cdot \gamma < 0$ and $\lambda \cdot \gamma > 0$. Hence $\operatorname{Re} G = u$. Moreover, $\widehat{G}(0) = 2\widehat{u}(0) - 1 = 1$, and every other frequency of G lies in the open half-space $\lambda \cdot \gamma < 0$. Sums of such frequencies stay in this half-space. Hence, for every $\varepsilon > 0$ and every polynomial A in one variable,

$$\int_{\mathbb{T}^d} A(G + \varepsilon)\,\mathrm{d}m = A(1 + \varepsilon). \tag{4}$$

Fix $\varepsilon > 0$ and write $w = G + \varepsilon$, so that $\operatorname{Re} w = u + \varepsilon \geq \varepsilon$. Choose $R > 1 + \varepsilon$ with $R > \|w\|_\infty^2/(2\varepsilon)$. Then

$$\left|1 - w/R\right|^2 = 1 - \frac{2\operatorname{Re} w}{R} + \frac{|w|^2}{R^2} \leq 1 - \frac{2\varepsilon}{R} + \frac{\|w\|_\infty^2}{R^2} < 1$$

uniformly on $\mathbb{T}^d$. Hence the principal logarithm has the uniformly convergent expansion

$$\log w = \log R - \sum_{k=1}^{\infty} \frac{(1 - w/R)^k}{k}.$$

The same expansion holds at the point $1 + \varepsilon$, since $0 < 1 - (1 + \varepsilon)/R < 1$. Multiplying by w, integrating term by term and using (4), we find that

$$\int_{\mathbb{T}^d} w \log w\,\mathrm{d}m = (1 + \varepsilon)\log(1 + \varepsilon).$$

We take real parts. Since $\operatorname{Re}(w \log w) = (u + \varepsilon)\log|w| - v \arg w$, $|w| \geq u + \varepsilon$ and $|\arg w| \leq \pi/2$, we get

$$\int_{\mathbb{T}^d} (u + \varepsilon)\log(u + \varepsilon)\,\mathrm{d}m \leq (1 + \varepsilon)\log(1 + \varepsilon) + \frac{\pi}{2}\|v\|_1.$$

Since u is bounded, we may let $\varepsilon \to 0$ by dominated convergence. This gives the first claim. For the second, note that $\operatorname{Im} P_\lambda u = v/2$, so that $\|P_\lambda u\|_1 \geq \|v\|_1/2$. □

4. The primorial example and the proof of Theorem 1

We now apply Lemma 1 to a product of d identical factors, one for each of the first d primes. Let $Q_d = \prod_{j \leq d} p_j$ be the dth primorial.

Lemma 2. *Suppose that $d \geq 1$, and let f_d be the Dirichlet polynomial with Bohr lift*

$$\mathcal{B}f_d(z) = 2^{-d} \prod_{j=1}^{d} (1 + z_j)^2.$$

Then $\|f_d\|_{\mathcal{H}^1} = 1$ and $\|S_{Q_d} f_d\|_{\mathcal{H}^1} \geq d(1 - \log 2)/\pi$.

Proof. Note that $(1+z)^2/2 = z(1+\cos\theta)$ for $z = e^{i\theta}$. Hence $\mathcal{B}f_d = z_1 \cdots z_d u_d$, where $u_d(z) = \prod_{j \leq d}(1 + \cos\theta_j)$. The function u_d is nonnegative, and Fubini's theorem gives $\int u_d\,\mathrm{d}m = 1$. Thus $\|f_d\|_{\mathcal{H}^1} = 1$.

The frequencies of u_d lie in $\{-1, 0, 1\}^d$, so the frequency γ of u_d corresponds to the term n^{-s} of f_d with $n = Q_d \prod_j p_j^{\gamma_j}$. Moreover $n \leq Q_d$ if and only

if $\lambda \cdot \gamma \le 0$. Hence $\mathcal{B}S_{Q_d} f_d = z_1 \cdots z_d P_\lambda u_d$, and so $\|S_{Q_d} f_d\|_{\mathcal{H}^1} = \|P_\lambda u_d\|_1$. By (2) we may apply Lemma 1, and it remains to compute the entropy of u_d. Since u_d is a product of probability densities in separate variables, Fubini's theorem gives

$$\int_{\mathbb{T}^d} u_d \log u_d \, \mathrm{d}m = d \int_{\mathbb{T}} (1+\cos\theta) \log(1+\cos\theta) \, \mathrm{d}m(\theta).$$

We claim that the last integral equals $1 - \log 2$. Writing $1+\cos\theta = 2\cos^2(\theta/2)$ and substituting $\theta = 2t$, we find that it equals

$$\log 2 + \frac{8}{\pi} \int_0^{\pi/2} \cos^2 t \log(\cos t) \, \mathrm{d}t.$$

Recall that $\int_0^{\pi/2} \log(\cos t) \, \mathrm{d}t = -(\pi/2)\log 2$. Integration by parts gives $\int_0^{\pi/2} \cos(2t) \log(\cos t) \, \mathrm{d}t = \int_0^{\pi/2} \sin^2 t \, \mathrm{d}t = \pi/4$. Since $\cos^2 t = (1+\cos 2t)/2$, the integral in the display equals $\pi/8 - (\pi/4)\log 2$, and the claim follows. Hence $\int u_d \log u_d \, \mathrm{d}m = d(1-\log 2)$, and Lemma 1 completes the proof. □

Lemma 2 gives the lower bound in Theorem 1 at the primorials $N = Q_d$. To reach every N, we multiply f_d by a suitable b^{-s}, which does not change the norm, and choose b such that the cutoff N acts on $b^{-s} f_d$ exactly as Q_d acts on f_d. This is the argument used in the proof of [1, Theorem 5.3], and it avoids any monotonicity assumption on $\|S_N\|_1$.

Proof of Theorem 1. The upper bound is [1, Theorem 5.2]. Since $S_N 1 = 1$, we have $\|S_N\|_1 \ge 1$ for every N. It is therefore enough to prove the lower bound for all sufficiently large N.

Let C_0 be the constant in (3). Set $d = \lfloor c_0 \log N / \log\log N \rfloor$, where the absolute constant $c_0 > 0$ is chosen so small that $4Q_d^2 \le N$ for all sufficiently large N; by (3) this holds for $c_0 < 1/(2C_0)$. Let $b = \lfloor N/Q_d \rfloor$. Then $b \ge N/Q_d - 1 \ge 4Q_d - 1 > Q_d$, and $N = bQ_d + r$ with $0 \le r < Q_d < b$. Hence $\lfloor N/b \rfloor = Q_d$.

Every term n^{-s} of f_d has $n \mid Q_d^2$, and the term $(bn)^{-s}$ of $b^{-s} f_d$ satisfies $bn \le N$ if and only if $n \le \lfloor N/b \rfloor = Q_d$. Consequently

$$S_N(b^{-s} f_d) = b^{-s} S_{Q_d} f_d.$$

Since multiplication by b^{-s} preserves the norm of $\mathcal{H}^1$, Lemma 2 gives

$$\|S_N\|_1 \ge \|S_{Q_d} f_d\|_{\mathcal{H}^1} \ge \frac{1-\log 2}{\pi} d \gtrsim \frac{\log N}{\log\log N}$$

for all sufficiently large N. □

5. Concluding remarks

The leading constant. The proof above uses only Chebyshev's bound. If we use the prime number theorem instead, then $\log Q_d \sim d \log d$ as $d \to \infty$.

Hence the largest d with $4Q_d^2 \le N$ satisfies $d \sim (1/2)\log N/\log\log N$. The proof of Theorem 1 then gives

$$\liminf_{N\to\infty} \frac{\log\log N}{\log N}\|S_N\|_1 \ge \frac{1-\log 2}{2\pi}.$$

The constant in the upper bound of [1, Theorem 5.2] is not explicit, and we have not attempted to make the two constants meet.

Problem 2. *Does the limit of $\|S_N\|_1 \log\log N/\log N$ as $N\to\infty$ exist? If so, determine it.*

Other values of p. The problem of Saksman and Seip also concerns $0<p<1$ and $p=\infty$. For $0<p<1$, Bondarenko, Brevig, Saksman and Seip [1, Theorem 5.3] showed that $\log\|S_N\|_p$ is of order $\log N/\log\log N$. For $p=\infty$, Chen and Fu [2, Theorem 1.1] recently proved that $\|S_N\|_\infty \ge (1/(2\pi)+o(1))\log N$, which together with an earlier upper bound of Balasubramanian, Calado and Queffélec (see [2, Section 1]) gives the order $\log N$. Our argument is specific to $p=1$, since Lemma 1 bounds an L^1 norm from below.

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